Evidence-Bounded Retrodiction (EBR)

Plain-English Concept + Mathematical Formalization for Review

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Based on the research paper: "From Memory to Causal History: Evidence-Bounded Retrodiction for Persistent Agents"

Purpose of this note: explain the idea without AI/research jargon, isolate the current mathematics, and give a mathematician a clean target for checking whether the definitions, invariants, ordering rules, and uncertainty treatment can be made more rigorous. This is not a claim that the mathematics below is a new theorem.

1. The idea in one sentence

When an AI reconstructs why something happened, it should keep separate (a) what a person was actually documented to know at the time, (b) what later evidence shows that person probably had access to at the time, and (c) everything we know now about what really happened. Mixing those three views creates hindsight errors.

2. A simple metaphor: the detective, the driver, and the dashcam

Imagine a car crash at 10:00 a.m. A driver makes a decision at 9:58 a.m. At noon, a detective investigates.

- View 1 - The driver's contemporaneous record: What messages, warnings, gauges, or reports were documented as available to the driver by 9:58?
- View 2 - Reconstructed historical access: At noon, we may discover that the driver actually saw a warning at 9:57, even though the server did not archive that message until 11:30.
- View 3 - Retrospective evidence: The detective can use a damaged brake part found at 11:20, even though the driver could not possibly have known about that inspection at 9:58.

The detective is allowed to use later evidence to explain the crash. But if we judge whether the driver's 9:58 decision was reasonable, we must not secretly give the driver information discovered at noon. EBR is a bookkeeping and reasoning protocol designed to enforce that separation.

3. Why ordinary timestamps are not enough

Most databases can tell us when an event happened and when the database recorded it. EBR argues that a persistent agent also needs an explicit notion of when a particular actor had access to the information.

Time concept	Meaning	Why it can fail
Event / valid time	When the real-world event or state occurred.	A past event can be discovered only later.
Record / transaction time	When the system stored the information.	A person may have seen information before the system archived it.
Actor-access time	Evidence that a particular actor had access by a particular time.	Must itself be supported by provenance; it is not mind-reading.

4. Mathematical objects

Let x be one evidence item. The paper attaches several pieces of metadata to x:

- I_v(x): the event/valid interval - when the underlying event or state belongs in world time.
- r(x): record time - when the system recorded the item.
- π(x): provenance - where the item came from and its source lineage.
- σ(x): epistemic status - e.g., documented, inferred, speculative, contradicted, or unknown.
- A(a,x,t): an actor-access assertion - evidence that actor a had access to item x by time t.

For a historical decision at time t_i and a later audit cutoff c (with c >= t_i), the paper uses two binary indicators:

- C(a,x,t_i) = 1 when access to x by t_i was contemporaneously attested by t_i.
- H_c(a,x,t_i) = 1 when evidence available by audit cutoff c supports that the actor had historical access to x by t_i.

5. The three core sets

$$K_{att}(a,t_i) = \{ x : r(x) \leq t_i \text{ and } C(a,x,t_i) = 1 \}$$

Plain English: the items that were already in the record by the decision time and were contemporaneously attested as accessible to the actor. This is the strongest 'what we can prove was in the actor's historical information state at that time' view.

$$K_{rec}(a,t_i ; c) = \{ x : r(x) \leq c \text{ and } H_c(a,x,t_i) = 1 \}$$

Plain English: using evidence available by the later audit time c, reconstruct what the actor had access to by the old decision time t_i. This may include an item archived after the decision if later provenance shows the actor had actually seen it earlier.

$$R(c) = \{ x : r(x) \leq c \} \cap S$$

Plain English: all evidence recorded by the audit cutoff c that falls inside the declared retrieval scope S. This is the detective's retrospective evidence set. It is not restricted to what the historical actor knew.

6. The key invariant: do not rewrite the past

The central mathematical/architectural constraint is non-retroactivity. Assume the transaction history is append-only: later records can be added, but they cannot be back-written to pretend they existed earlier.

$$\text{Later insertion after } t_i \Rightarrow K_{att}(a,t_i) \text{ does not change}$$

However, later evidence may legitimately expand $K_{rec}(a,t_i;c)$ or $R(c)$ as c moves forward. In metaphor: the detective can learn more, and can even discover what the driver had seen, but cannot edit the driver's contemporaneous file so that the noon discovery appears to have been documented at 9:58.

7. Candidate causal histories

EBR then works backward from a present observation and constructs candidate histories. A candidate history h is not accepted merely because its events occurred in the right order or look similar to known patterns. Required causal links must have explicit mechanism bridges.

For a proposed edge u -> v, the system should ask: What mechanism connects u to v? What independent evidence supports that mechanism? What intermediate consequence should we expect? Is there contradictory evidence? Is the link documented, inferred, speculative, or unknown?

8. The evidence profile for a candidate history

$$\Phi(h) = < m(h), n_{\pi}(h), c(h), u(h) >$$

The four components are:

- m(h): supported / required mechanism-bridge coverage. Example: if a history needs 3 causal bridges and all 3 are supported, coverage is 3/3.
- $n_{\pi}(h)$: number of independent provenance roots after deduplicating sources that ultimately come from the same origin.
- c(h): burden of adjudicated contradictions under the relevant domain evidence policy.
- u(h): number of unresolved required mechanism bridges.

Important: Φ(h) is deliberately NOT a probability vector and is not collapsed into one weighted score. The current paper avoids arbitrary weights and avoids forcing incomplete candidate histories to sum to 100%.

9. Admissibility and ordering

The prototype uses a non-compensatory rule. A missing required causal bridge cannot be 'paid for' by having lots of weaker supporting evidence. Likewise, an adjudicated hard contradiction cannot simply be outweighed by many correlated reports.

$$\text{Admissible}(h) = 1 \text{ only if } u(h) = 0 \text{ and } c_{\text{hard}}(h) = 0$$

The executable prototype then uses a lexicographic display ordering roughly equivalent to:

$$\text{RankKey}(h) = (\text{Admissible}(h), -c(h), \text{coverage}(h), n_{\pi}(h), -u(h))$$

This is a display/ranking convention, not a statistical posterior. The mathematician may want to replace this implementation convenience with a cleaner partial order or dominance relation that preserves incomparability.

10. Open-set uncertainty: the missing-story reserve

Suppose we have considered three explanations, H1, H2, and H3. Ordinary software is often tempted to normalize those three and say, for example, 70%, 20%, 10%. But what if the real explanation is H4 and nobody thought of it?

$$\perp = \text{unmodeled-history reserve}$$

The symbol ⊥ represents the fact that the enumerated hypotheses are not assumed exhaustive. It is not assigned an arbitrary percentage in the current formulation. Because ⊥ remains active, the system may say 'H1 is strongest among the admissible histories we examined' rather than 'H1 is the true cause with 70% probability.' It may also abstain.

11. Worked example from the prototype

The toy system is an automated production cell. Power is lost at 10:01. An operator makes a response decision at 10:05. The audit occurs at 12:00.

- A grid warning was seen by the operator at 09:57, but the archive containing that warning was not imported until 11:30.
- A post-event inspection at 11:20 reports damage associated with a 09:56 disturbance. The inspection plainly was not available to the operator at 10:05.
- The three candidate histories are: H1 external grid disturbance, H2 internal overload, H3 manual shutdown.

Historical view	Leakage	Actor precision	Actor recall
Valid-time only	1 item	0.83	1.00
Record-time only	0	1.00	0.80
Three-view K_rec	0	1.00	1.00

Why? Valid-time-only reasoning leaks the later inspection into the operator's past information state because the inspection concerns an earlier event. Record-time-only reasoning misses the grid

warning because the archive arrived late. Actor-access reconstruction handles both timing cases correctly in this deterministic toy example.

At audit time, H1 has all three required mechanism bridges supported by four independent provenance roots and no contradiction. H2 has an unresolved bridge. H3 has some supporting chat evidence but is contradicted by an authentication log. The prototype therefore reports H1 as the strongest admissible candidate among the enumerated histories while keeping the unmodeled-history reserve active.

12. Precision and recall in the toy test

$$\text{Precision} = \text{true-access items retrieved} / \text{all items retrieved}$$

$$\text{Recall} = \text{true-access items retrieved} / \text{all true-access items}$$

These metrics are used only because the synthetic world has known access labels. A real-world implementation would need reliable ground truth or adjudicated access records before these numbers could be interpreted as performance evidence.

13. What the mathematics is NOT claiming

- It is not claiming that chronology alone proves causation.
- It is not claiming that $\Phi(h)$ is a probability distribution.
- It is not claiming a new causal-identification theorem.
- It is not claiming that the unmodeled reserve can recover an explanation for which no evidence exists.
- It is not claiming that later reconstruction of actor access is mind-reading; it is an evidential assertion that must carry provenance.
- It is not claiming real-world causal accuracy from the toy prototype.

14. Where I want mathematical scrutiny

The following are the main questions I would like a mathematician to examine. The goal is not to make the paper look more mathematical; the goal is to find the cleanest formal structure that actually matches the intended semantics.

Area	Question
Set semantics and time	Are K_att, K_rec, and R defined cleanly enough? Should actor-access be represented as a relation, event, interval, filtration, or another structure?
Non-retroactivity	Can the append-only condition and invariance of K_att be stated as a precise proposition with minimal assumptions? What monotonicity properties should K_rec and R have as audit

	cutoff c increases?
Evidence provenance	What is the right mathematical way to count or compare 'independent provenance roots' without assuming independence that has not been demonstrated?
Contradictions	Should contradictions be represented through paraconsistent logic, signed evidence, a lattice, constraints, or domain-specific adjudication rather than a simple count?
Mechanism bridges	Can admissibility be formalized as graph constraints or proof obligations, especially when a bridge can be documented, inferred, speculative, or unknown?
Ordering candidate histories	Would a partial order / Pareto dominance relation be more defensible than the prototype's lexicographic RankKey? How should ties and incomparable histories be handled?
Open-set reserve	What is the cleanest formal meaning of ⊥ when hypotheses are non-exhaustive but we intentionally refuse to assign unvalidated probabilities?
Probabilistic extension	If a future domain supplies calibrated likelihoods or a valid structural causal model, how could Bayesian or causal probabilities be added without destroying the three-view separation?
Falsifiability	Which mathematical properties should be tested independently of empirical causal accuracy: invariance, leakage prevention, monotonicity, contradiction preservation, and abstention consistency?

15. A possible stronger formal direction

One possible route, subject to mathematical review, is to view the historical information available to an actor as a time-indexed family of provenance-bearing sets, while treating retrospective audit evidence as a different filtration or evidence process. The core prohibition would be that later audit information may update beliefs about historical access and causal history but may not mutate the contemporaneously attested state. Candidate histories could then live in a constrained causal graph space with a partial order induced by mechanism completeness, contradiction status, and provenance support. This paragraph is a proposed formal direction, not a claim already established by the paper.

16. The conceptual payoff

The larger aim is to let a persistent AI learn from history without rewriting history. The same system should be able to say both: 'Knowing everything we know now, this is the strongest supported causal reconstruction,' and 'Given only what the actor could reasonably have known then, that historical decision was or was not justified.' Those are different questions. EBR's mathematics is intended to keep them different.

17. One-paragraph request to the mathematician

I would value a rigorous review of the definitions and invariants above, especially the separation of contemporaneous knowledge, reconstructed historical access, and retrospective evidence; the formal treatment of non-retroactivity; the evidence-profile ordering; and the open-set reserve. Please identify any hidden assumptions, inconsistent notation, structures that should be replaced with more standard mathematics, and any propositions that can be proved without overstating what the method establishes. I am specifically trying to avoid decorative mathematics, arbitrary weights, and pseudo-probabilities.

Source note

This review note is derived from the September 2026 manuscript “From Memory to Causal History: Evidence-Bounded Retrodiction for Persistent Agents.” The manuscript positions EBR as an architectural integration/protocol and evaluation target, not as a new general theory of causality, temporal databases, provenance, abduction, or causal identification.